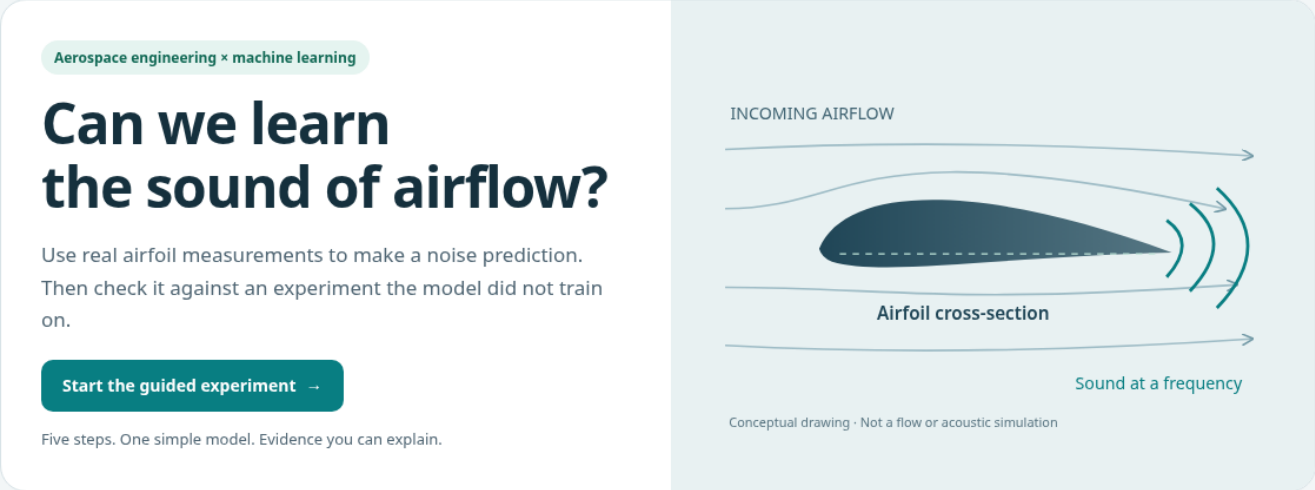
AEROSPACE ENGINEERING × AI & DATA SCIENCE

AeroNoise AI

Student guide & research-pitch playbook

Research Pitch Day · Aerospace Engineering
Noorul Islam Centre for Higher Education, Kanyakumari



Your task: explain a small aerospace problem, run a simple machine-learning comparison, and show whether the prediction agrees with held-out experimental measurements.

Open the app

Standalone download: save **AeroNoise_AI.html**, then open it in a desktop browser. **Repository ZIP:** extract it first, then open **index.html**. The two filenames contain the same app. On Windows, use right-click → Open with → Chrome or Edge when needed.

Everything needed for the calculations is inside the HTML file. No Python, terminal, API key, paid AI model, or data upload is needed. JavaScript must be enabled. External source links need internet access.

The learning route

1 Understand → 2 Explore data → 3 Train & test → 4 Try a case → 5 Pitch & export

Begin with Experiment A. Leave the high-speed challenge until the team can explain training, testing, the baseline, and prediction error. Test the app on the presentation laptop before the event.

01 / LEARN THE PROBLEM

What are we predicting?

An **airfoil** is a wing-like cross-sectional shape. Air flowing over it can generate noise. This app uses a public NASA dataset, distributed by UCI, rather than experiments carried out by your team. The source contains **1,503 observations** of NACA 0012 airfoils and five inputs. [1]

Input	Unit	What to say
Frequency	Hz	The sound frequency we are examining.
Angle of attack	degrees	The angle between the airfoil chord and incoming airflow.
Chord length	m	The leading-edge to trailing-edge distance.
Free-stream speed	m/s	The incoming airflow speed.
Displacement thickness	m	A suction-side boundary-layer parameter supplied in the dataset; not wing thickness.

The output is scaled sound pressure level (dB). Keep the word “scaled”. This is not the total sound of an aircraft, drone, or airport. The app does not redesign a wing or certify noise limits.

The two prediction methods

Baseline: calculate the training-set average noise target and predict that same value for every test case.

Five-neighbour regression: make the five input scales comparable, find the five closest training observations, and average their measured target values. This is supervised machine learning, not a neural network. [2]

Why separate training and testing?

A model must be checked on observations it did not use for fitting. Here, identical angle, chord, speed, and displacement-thickness combinations form **106 condition groups**. Frequencies within each group stay together. These groups are constructed by the app, not supplied experimental run IDs. [3, 4]

The input means and standard deviations come only from training rows. Using test rows to fit this scaling would make the evaluation less trustworthy. [3]

02 / RUN THE DEFAULT EXPERIMENT

Experiment A: new conditions

Action	What you should see or do
Open Step 3	Keep “A · New conditions” selected.
Check the split	1,188 training rows / 84 groups; 315 test rows / 22 groups; no group overlap.
Click “Train and test”	Wait for the actual calculation to finish. The model settings are fixed at five neighbours.
Read the results	Compare model MAE with baseline MAE on the same 315 test rows.
Read the plot	Dots near the diagonal have close agreement between predicted and measured values.

KNN AVERAGE ABSOLUTE ERROR

3.222 dB

Mean absolute error (MAE). Lower is better on this same test set.

BASELINE AVERAGE ABSOLUTE ERROR

5.201 dB

The training-mean predictor evaluated on identical test observations.

RELATIVE MAE REDUCTION

38.1%

Reduction in prediction error versus the baseline. **Not accuracy or reduced aircraft noise.**

Metric	Five-neighbour model	Average baseline
MAE, dB	3.222	5.201
RMSE, dB	4.450	6.335
R^2	0.506	−0.002

These are the actual calculated version 1.0.0 results, checked independently in Python. Small displayed rounding differences are normal. Re-running the unchanged app gives the same result.

Explain MAE without jargon

“Across the 315 test observations, the model’s average absolute prediction error is about **3.22 dB**. The baseline’s error is about **5.20 dB**. The model therefore does better in this particular test.”

Do not say “38% accurate”. The displayed 38.1% is the relative reduction in MAE versus the baseline. It is not accuracy and does not mean an aircraft became 38.1% quieter.

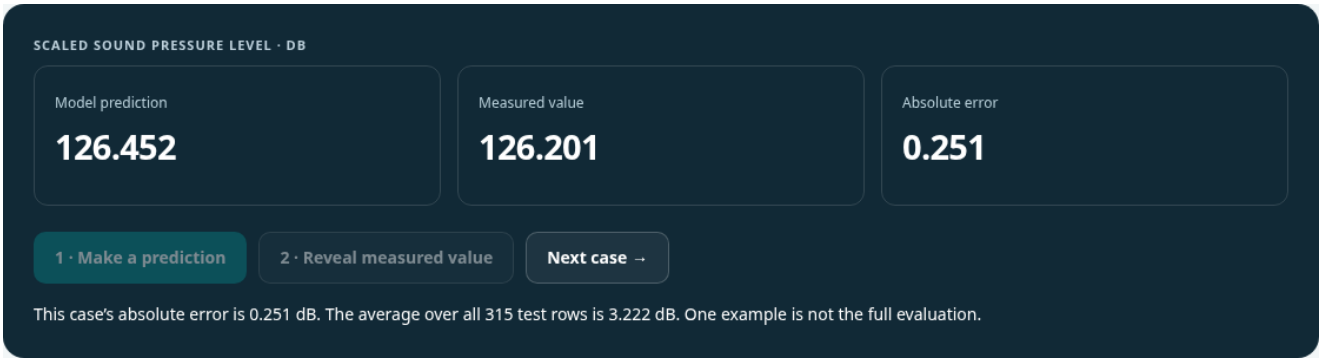
RMSE gives larger errors extra weight. R^2 compares squared error with a test-mean reference and can be negative; it is not percentage accuracy. The main result students should understand is MAE.

03 / DEMONSTRATE THE PREDICTION

Predict, then reveal

After Experiment A finishes, select **Step 4 · Try a case**. The app fills in the five measured inputs for a held-out observation. Do not invent a value for displacement thickness.

Button	Demonstration instruction
1 · Make a prediction	Read the model estimate before revealing the measurement.
2 · Reveal measured value	Compare the experimental value and the absolute error.
Show how the five neighbours…	Point out that all five neighbours came from training rows.
Next case	The next held-out row loads and the measurement is hidden again.



The first Experiment A case is original row 1. The displayed prediction is 126.452 dB, the measurement is 126.201 dB, and the absolute error is 0.251 dB. This is an illustration, not a replacement for the 315-row test evaluation.

What to tell the audience

> “This case was held out from training. The model found five similar training observations and averaged their target values. We then revealed the real measurement to check the error. A single good case is not proof that every prediction will be good.”

One important detail

The app averages numerical target values as a regression estimate. It is **not** adding the acoustic energy of five sound sources. The neighbour distance is a standardised mathematical distance, not a physical separation or confidence score.

A pitch everyone can explain

Team role	Main responsibility
Member 1	Introduce the aerospace problem, the target, and the research question.
Member 2	Explain the five inputs, condition groups, and training/test separation.
Member 3	Explain the baseline and five-neighbour method.
Member 4	Operate the case demonstration, state the actual error, and explain the limitations.

Suggested speaking script

“Our project is AeroNoise AI, a browser-based study of airfoil-noise prediction. We use public NASA wind-tunnel measurements from UCI. Five numerical inputs are used to estimate scaled sound pressure level at a frequency. We compare five-neighbour regression with a training-average baseline. In Experiment A, we train on 1,188 observations and test on 315 held-out observations. The model MAE is 3.22 dB, compared with 5.20 dB for the baseline. Our contribution is an accessible, reproducible demonstration and a clear evaluation question, not a new machine-learning algorithm. This is a dataset-specific result, not a certified aircraft design or safety tool.”

Prepare in four short sessions

Understand: explain each input and target in your own words. **Operate:** run Experiment A and reveal several cases. **Interpret:** explain why MAE matters and what the result does not establish. **Rehearse:** rotate the speaking roles so every member knows the whole workflow.

Before the event, each student should answer

What are the inputs and output? Which rows train the model? What does the baseline predict? Why are there five neighbours? What is a held-out case? Why is the result not “percentage accuracy”?

Acknowledge assistance. The app and guide were prepared with AI assistance. Students should not claim to have written every line independently or collected the NASA measurements. Credit the data source and explain the work they actually understand and reproduce.

05 / EXTEND CAREFULLY

The high-speed research challenge

After mastering A, return to Step 3, choose **B · Unseen high speed**, and run it. This trains at 31.7, 39.6, and 55.5 m/s and tests all rows at 71.3 m/s, outside the training speed range.

Quantity	Experiment B result
Training / test rows	1,038 / 465
Training / test groups	73 / 33; no overlap
Model / baseline MAE	2.384 / 5.951 dB
Model RMSE / R^2	3.291 dB / 0.779

B happens to have lower error than A here. The test populations differ. Do not conclude that extrapolation is generally easier or reliable. Compare each model only against its own baseline, and treat this as one fixed evaluation.

How to export the evidence

Step 5 download	How to use it
Printable report	Open the downloaded HTML file, choose Print, and select Save as PDF. The appendix contains all test predictions.
Test predictions CSV	Open in a spreadsheet to inspect measured values, predictions, and errors.
Methods + results JSON	Preserve the exact split IDs, settings, scaling, full-precision outputs, and source details.
Pitch text	Use as rehearsal material, then present in your own words.

The selected experiment determines the exported evidence. An optional team name stays in the browser until it is placed in a downloaded report. Refreshing or closing the app clears the current session; downloaded files remain on the device.

Share the app through GitHub Pages

1. Create a public repository, for example **aero-noise-ai**. **2.** Extract the supplied ZIP and upload its contents. **index.html must be at the top level.** Do not upload only the ZIP. **3.** Commit the files to the main branch.

4. Open **Settings → Pages**. Select **Deploy from a branch → main → /(root) → Save**. **5.** Open the website address GitHub displays after successful deployment. Share that address, not the repository page showing the code. GitHub Pages supports public repositories on GitHub Free. [5]

No live site has been published with this package. Publication is the separate step above. Do not upload private documents or students' personal information. Keep the local HTML file as a presentation backup.

Quick troubleshooting

You see code: open the HTML file in a browser, not a text editor; on GitHub use the Pages website. **Pages shows 404:** check deployment status, the selected branch/root, and the exact filename index.html. **No predictions:** enable JavaScript and run Step 3 first. **Copy fails:** use Download pitch as text instead. **School device blocks local HTML:** use the published page or ask the authorised administrator; do not bypass security restrictions.

What has been checked?

The complete source file matches a published checksum. All 780 predictions across A and B match an independent NumPy calculation. Offline Chromium tests passed for navigation, calculation, prediction/reveal, downloads, quiz/reset, and 360/390 px layouts, with no JavaScript errors or network requests.

Testing limitation: this environment blocks file:// and localhost navigation. Tests rendered the actual HTML directly in an offline Chromium context. Double-click navigation, a live GitHub Pages site, and other browser engines were not tested here. Rehearse on the students' intended device. Full methods and test logs are in the repository README and verification folder.

Sources and attribution

[1] Brooks, T.; Pope, D.; Marcolini, M. (1989). Airfoil Self-Noise. UCI Machine Learning Repository. DOI: 10.24432/C5VW2C. Dataset: CC BY 4.0; numerical values unchanged. Row/group IDs and model outputs added by this app.

[2] Scikit-learn documentation: Nearest-neighbour regression.

[3] Scikit-learn documentation: Common pitfalls and preprocessing leakage.

[4] Scikit-learn documentation: Cross-validation and grouped evaluation.

[5] GitHub documentation: Configuring a publishing source for GitHub Pages.

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